

Five reproducible finance analytics projects

A portfolio built on **Python + DuckDB** pipelines over free, adversarially-verified data sources — from a formal COVID event study to a vulnerability-adjusted vendor scorecard. Every number traceable to a primary-source filing, government record, or pinned public dataset. Project 5 is **built & runnable**.

PROJECT 01

Post-COVID Sector Rotation

Event study & macro linkage across the 11 U.S. equity sectors, 2019–2025.

PROJECT 02

Luxury Goods Comps

Market share, margins & valuation multiples across the listed global luxury houses.

PROJECT 03

Fortune 500 Breakdowns

A multi-year panel of corporate America — profitability, efficiency, churn.

PROJECT 04

Cloud, ERP & AI Vendor Intelligence

Segment revenue vs. security-risk exposure — finance × cybersecurity.

PROJECT 05

Industry Value-Chain

Margin-distortion across 94 U.S. industries — Damodaran datasets → Supabase.

- Python
- DuckDB
- Parquet caching
- pandera quality gates
- Streamlit dashboards
- pytest
- GitHub Actions CI

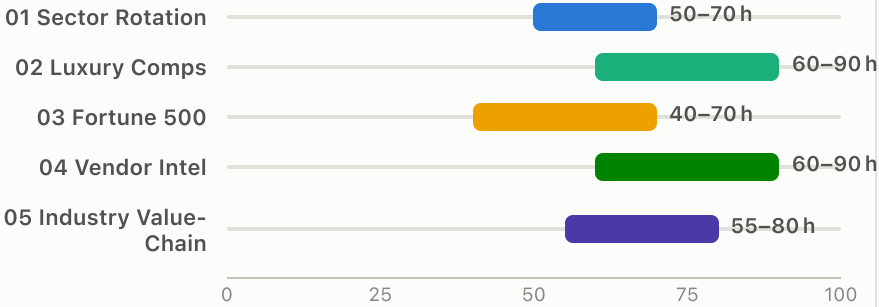
Five projects, one engineering spine

Each project stands alone, but they share a reusable toolkit — a cached ingestion layer, a layered DuckDB warehouse, quality gates, and CI — so effort compounds across the sequence.

THE LINEUP

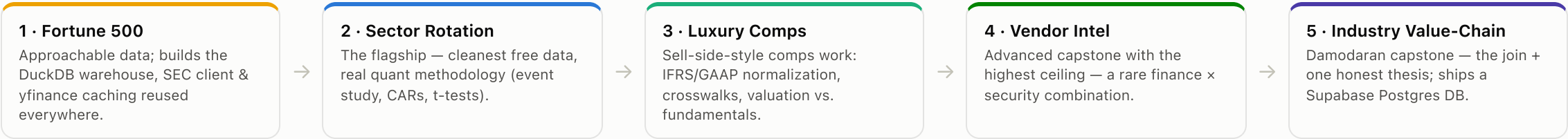
	ANGLE	DIFFICULTY	EFFORT	DATA	ROLE
01 Sector Rotation	Event study + macro linkage	Interm.→Adv.	~50–70 h	★★★★★	Flagship — best effort-to-impressiveness
02 Luxury Comps	Comps, multiples, market share	Interm.→Adv.	~60–90 h	★★★★★	Most finance-recruiter appeal
03 Fortune 500	Panel data engineering	Intermediate	~40–70 h	★★★★★	Quick win — builds the shared toolkit
04 Vendor Intel	Finance x cybersecurity	Advanced	~60–90 h	★★★★★	Capstone — most differentiated
05 Industry Value-Chain	Entity resolution, margins vs valuation	Advanced	~55–80 h	★★★★★	Built — runnable + Postgres DB

ESTIMATED EFFORT (HOURS)



Range = documented estimate per project brief. Luxury and Vendor Intel run longer because PDF extraction and CVE attribution are manual-judgment work.

RECOMMENDED BUILD ORDER



Post-COVID Sector Rotation

An event-study and macro-linkage analysis of U.S. equity sectors, 2019–2025 — how the 11 sectors crashed, recovered, and structurally re-ranked through COVID-19.

DIFFICULTY

Interm. → Adv.

Real quant methodology

EFFORT

~50–70 h

Best effort-to-impressiveness ratio

DATA CLEANLINESS

★★★★★

Fully automatable, no PDFs

UNIVERSE

11 ETFs + SPY

XLK, XLF, XLE, XLV, XLY, XLP, XLI, XLU, XLB, XLRE, XLC

WINDOW

2019 → 2025

Daily prices · 8 macro series · stringency

THE INTERVIEW STORY

"I ran a formal **market-model event study** of the COVID crash across all 11 GICS sectors, measured **cumulative abnormal returns vs. the S&P 500** over a 250-day estimation window, and showed which sectors were structural losers vs. winners — then linked recovery speed to sector-specific macro damage and the pandemic stringency timeline."

THREE ANALYTICAL LAYERS

1 · Descriptive

Returns, rolling vol, max drawdown, recovery time, Sharpe/Sortino, correlation regimes.

2 · Event study

CARs vs. market-model expectation around 19 Feb → 23 Mar 2020; t-tested significance.

3 · Macro linkage

FRED macro + daily VIX + stringency; regress recovery speed on macro damage.

KEY ANALYTICAL QUESTIONS

- 1 Which sector had the **deepest drawdown** — and does that ranking match the ranking by **recovery time**?
- 2 Which sectors show statistically significant negative CARs (Energy, Financials?) vs. positive (Tech, Staples?) — and does the sign **flip in the 2021–22 inflation regime**?
- 3 Did cross-sector correlations **spike toward 1.0** in the March-2020 panic — "diversification fails when you need it"?
- 4 Which sectors carried the highest **VIX beta** during the shock, and how long did volatility take to normalize?
- 5 Comparing 2019 vs. 2023–25 leadership: which sectors **permanently re-ranked** vs. mean-reverted?

Sector Rotation — pipeline, data, risks

■ VERIFIED DATA SOURCES

SOURCE	ROLE	ACCESS
yfinance / Stooq readers	Primary price panel — returns, vol, drawdowns, abnormal returns	No key; fetch once, cache to Parquet
FRED	VIXCLS, UNRATE, INDPRO, CPI, sector payrolls — macro linkage & VIX beta	Free 32-char API key
Our World in Data	COVID stringency + case/death waves overlay	⚠ Frozen 19 Aug 2024 — scope overlay to ≤2024
Oxford OxCGRT	Containment vs. economic-support sub-indices (optional)	Direct CSV, no key
BLS / BEA	Industry payrolls; GDP-by-industry “markets lead the economy” (optional)	Free instant keys

■ FLAGSHIP VISUALS

- Drawdown vs. days-to-recover ranking
- CAR paths, significance-flagged
- Correlation-regime heatmaps
- Rolling-vol small multiples
- VIX-beta ranked bar
- 2019 vs. 2025 leadership bump chart

■ METHODOLOGY

- 1 Ingest once, cache to Parquet — prices, FRED, OWID; retry/back-off.
- 2 Land raw in DuckDB; staging computes returns, aligns calendars.
- 3 Descriptive marts — vol, drawdown, recovery, Sharpe, correlations.
- 4 Event study — per-sector α , β over 250 pre-event days; CARs + t-tests.
- 5 Macro linkage — explicit resampling; VIX betas; recovery regressions.
- 6 pandera gates → figures → Streamlit → pytest + CI → findings README.

■ RISKS & GOTCHAS

- Stooq CSV sits behind a JS anti-bot wall — use library fetchers + cache, never naive requests.
- yfinance is fragile (HTTP 429) — fetch once, cache, retry; never depend on it live.
- Frequency mismatch (daily / monthly / quarterly) is the main analytical trap — resample explicitly.
- Event-study validity needs a **clean 2019 estimation window**; frame macro results as associations, not causation.

Luxury Goods Market Share & Valuation

A comps model benchmarking the listed global luxury houses — LVMH, Kering, Richemont, Hermès, Burberry, Prada, Brunello Cucinelli, Estée Lauder, Tapestry, Capri — on share, mix, margins, and multiples.

DIFFICULTY

Interm. → Adv.

Real sell-side analyst work

EFFORT

~60–90 h

PDF extraction is the slow part

DATA CLEANLINESS

★★★★★

Filings PDFs need hand-mapping

UNIVERSE

Listed peer set

IFRS + US GAAP · EUR/CHF/GBP/USD

WINDOW

FY2015 → 2024

Annual, from primary filings

■ THE INTERVIEW STORY

"I built a comps model across the listed global luxury houses — normalizing **IFRS vs. US-GAAP** filings and four currencies, hand-mapping each house's segment and geographic taxonomy into a common schema, then benchmarking margins and **EV/EBITDA, EV/Sales and P/E vs. Damodaran** industry averages to flag who's expensive for their fundamentals. Hermès's structural margin premium shows up cleanly."

■ FOUR PILLARS

Market share & HHI

Revenue share *within the listed peer set* — reproducible from primary filings.

Revenue mix

Segment & geography, hand-mapped to a common schema across houses.

Financial performance

3/5/10-yr CAGRs, gross/op/net margins, currency-normalized to EUR & USD.

Valuation

EV/Sales, EV/EBITDA, P/E vs. Damodaran; valuation-vs-fundamentals scatters.

■ KEY ANALYTICAL QUESTIONS

- 1 How has **peer-set HHI** evolved 2015–2024 — did the 2024 downturn consolidate or fragment share?
- 2 Which **segment mix** proved most resilient through the 2024 contraction?
- 3 Geographic exposure: who is most exposed to the **China slowdown** vs. Japan/US strength?
- 4 Does **Hermès's structural margin premium** show cleanly vs. Damodaran's apparel average?
- 5 Do multiples rank houses the same way growth and margins do — **who's expensive for their fundamentals**?
- 6 Where do yfinance fundamentals **diverge from SEC EDGAR** ground truth for the US filers?

Luxury Comps — pipeline, data, risks

■ VERIFIED DATA SOURCES

SOURCE	ROLE	ACCESS
SEC EDGAR XBRL	Ground-truth financials for US filers (EL, TPR, CPRI); reconciliation benchmark	No key; User-Agent header; 10 req/s
Company URDs / annual reports	Primary segment & geographic revenue for all houses (IFRS filers absent from EDGAR)	⚠ PDFs; taxonomies differ → crosswalk table
yfinance	Prices, market cap, EV inputs for multiples	⚠ Non-US fundamentals often empty — prices only
Damodaran (NYU Stern)	Industry-average margins & multiples benchmark	⚠ URLs overwritten each January — pin file + date
Bain × Altagamma	Total-market denominator & narrative context	Free headline figures (full decks gated)

■ FLAGSHIP VISUALS

- Market-share explorer + HHI trend
- Segment-mix stacked bars
- Geographic-exposure view
- Margin ladder w/ Damodaran markers
- Valuation-vs-fundamentals scatter
- yfinance-vs-EDGAR variance panel

■ METHODOLOGY

- 1 Ingest EDGAR XBRL, URD PDFs, yfinance prices, Damodaran — cache all raw.
- 2 **Hand-map each house’s segment & geo taxonomy** into a common schema — a documented crosswalk, itself a defensible artifact.
- 3 **Currency-normalize** to EUR & USD; align March fiscal year-ends (Richemont, Burberry).
- 4 Marts: revenue share, HHI, CAGRs, margins, EV/Sales, EV/EBITDA, P/E.
- 5 Benchmark vs. Damodaran; **reconcile yfinance vs. EDGAR** and quantify variance.
- 6 pandera gates → multi-page Streamlit → pytest over every formula → CI.

■ RISKS & GOTCHAS

- **Taxonomies differ materially** (LVMH groups vs. Kering tiers vs. Richemont maisons) — treat the crosswalk as a transparent, first-class artifact.
- **EDGAR covers only US filers** — European/Asian financials come from PDFs (manual).
- FX convention (spot vs. average) is an **analyst choice — document it**.
- Don’t scrape Statista/Macrotrends; note corporate actions (Prada–Versace, Capri/Tapestry) so comps aren’t distorted.

Fortune 500 Financial Breakdowns

A multi-year DuckDB panel of corporate America's largest firms — sector profitability, revenue-per-employee efficiency, geographic concentration, and year-over-year list churn.

DIFFICULTY

Intermediate

Best pure data-engineering showcase

EFFORT

~40–70 h

Most approachable data

DATA CLEANLINESS

★★★★★

Community CSVs

PANEL SIZE

~5,000 rows

~10 years × 500 companies

DELIVERABLE

make pipeline

Reproducible on a fresh clone, free data only

THE INTERVIEW STORY

"I engineered a reproducible **DuckDB warehouse over ~10 years of Fortune 500 data**, then showed that tech and pharma dominate margins while retail dominates headcount, that **revenue-per-employee varies ~20× across sectors**, and that ~3–5% of the list turns over annually with churn concentrated in energy and retail."

CORE DELIVERABLE = THE DATA ENGINEERING

- Canonical schema at company-year grain: rank, revenue, profit, assets, employees, sector, HQ + derived **net margin, revenue-per-employee, rank change**.
- **Cross-year entity resolution** — track firms through rebrands and M&A via a documented alias/override table.
- Fact/dimension model + **SQL window functions** for churn flags and rank movement.
- Quality gates: exactly 500 ranks/year, non-null keys, non-negative financials.

KEY ANALYTICAL QUESTIONS

- 1 **Does size predict profitability?** Or does net margin decouple from scale, overall and by sector?
- 2 Which sectors are most/least profitable (median margin, ROA) — and how have rankings shifted YoY?
- 3 How does **revenue-per-employee** vary — which industries are the ~20× outliers?
- 4 How concentrated is the list (top-10/25/50 share, HHI) and **where is corporate America headquartered?**
- 5 **How churny is the list?** Entry/exit rates, sector churn as a disruption proxy, survival of former top-100 firms.

Fortune 500 — pipeline, data, risks

■ VERIFIED DATA SOURCES

SOURCE	ROLE	ACCESS
cmusam/fortune500 (GitHub)	Longitudinal spine → base fact table, company-year grain	△ Per-year files 2005–2019 (combined CSV 404s) — pin commit SHA, mirror raw
Kaggle datasets	Enrichment: sector, industry, HQ, employees, market cap	Free token; prefer CC0/CC BY; pin slug + version
Wikipedia	Current-year refresh & cross-check (~top 100)	pandas.read_html, set a User-Agent
SEC EDGAR XBRL	Validation layer: audited revenue/income/assets for public filers	No key; User-Agent header required
Damodaran (NYU Stern)	Industry benchmark margins/ROC for context	Direct .xls; taxonomy mapping needed

■ FLAGSHIP VISUALS

- Size-vs-margin scatter
- Sector profitability boxplots
- Revenue-per-employee ranked bars
- Concentration & HHI trends
- US-state HQ choropleth
- Churn stacked bars + rank bump chart

■ METHODOLOGY

- 1 Typed config with **pinned SHAs/versions**; ingest and cache all raw sources.
- 2 Land raw verbatim in DuckDB with source + ingestion lineage.
- 3 Normalize names, types, **units to USD millions**, sector labels, HQ fields.
- 4 **Cross-year entity resolution** → stable company_id via alias table.
- 5 Marts: fact_company_year + dim_company; window functions for churn & rank change.
- 6 Quality gates → 8 analyses → Streamlit + pytest + CI + data dictionary.

■ RISKS & GOTCHAS

- Fortune rankings are **proprietary editorial data** — community CSVs are reproductions; cite them as such.
- **URLs drift/404** — pin commit SHA / dataset version and mirror raw files into the repo.
- Employees/sector/HQ columns are **inconsistent across years**; fiscal conventions won't tie exactly to SEC.
- Nominal dollars — **deflate with CPI** before making trend claims.

Cloud, ERP & AI Vendor Intelligence

Segment revenue, implied market position, and security-risk exposure — fusing SEC filings, developer-adoption signals, and the NVD/CISA vulnerability record across ~10 enterprise-tech vendors.

DIFFICULTY

Advanced

Highest ceiling, most differentiated

EFFORT

~60–90 h

CPE attribution is the hard part

DATA CLEANLINESS

★★★★★

Entity resolution across 3 ID systems

UNIVERSE

~10 vendors

MSFT, AMZN, GOOGL, ORCL, CRM, SAP, NOW, SNOW (+IBM, WDAY)

WINDOW

FY2016 → now

Filings + 10k+ CVEs + KEV catalog

■ THE INTERVIEW STORY

“The enterprise-tech vendors compounding revenue fastest are also accumulating the **most severe, most-exploited vulnerabilities** — here’s that trade-off, measured, with every number traceable to a primary-source filing or a government vulnerability record.”

■ THREE PILLARS + A CAPSTONE HYPOTHESIS

A · Financials

Reportable-segment revenue, operating income, YoY growth from SEC EDGAR XBRL.

B · Market position

Revenue mix within the peer set + free adoption signals — *never* fabricated market share.

C · Security exposure

NVD CVE burden, CVSS mix, CISA KEV exploited-rate, publication-to-KEV lag.

Capstone: join A × C into a vendor scorecard and test one crisp hypothesis — is higher cloud-revenue growth associated (Spearman) with a heavier severe/exploited vulnerability load?

■ KEY ANALYTICAL QUESTIONS

- 1 How does **AWS’s clean segment disclosure** compare to MSFT’s blended “Intelligent Cloud” and GCP’s line — where does analysis hit the disclosure wall?
- 2 Do **developer-adoption signals** (Stack Overflow survey, GH Archive) corroborate the revenue momentum ranking?
- 3 Which vendors carry the largest **CVE burden**, and how has their severity mix shifted YoY?
- 4 What fraction of each vendor’s CVEs became **actively exploited** (CISA KEV) — and how fast?
- 5 Is faster cloud growth **statistically associated** with heavier severe/exploited load?

Vendor Intelligence — pipeline, data, risks

■ VERIFIED DATA SOURCES

SOURCE	ROLE	ACCESS
NVD CVE API 2.0	Primary security dataset — CVEs per vendor via CPE matching; CVSS severity + dates	Free key → 50 req/30s; paginate, cache
CISA KEV catalog	Actively-exploited flags → exploited-rate, publication-to-KEV lag	JSON/CSV, no key; CCO mirror
SEC EDGAR XBRL	Primary financial dataset — total & segment revenue/income per vendor	⚠ Segment tags non-standard — may parse 10-K notes
Stack Overflow survey	Adoption corroboration: YoY tool-usage share	⚠ Self-selected; columns change yearly — directional only
GH Archive / yfinance / MITRE	Optional: mindshare proxy · equity panel · bulk CVE fallback	BigQuery free tier · cache · shallow clone

■ FLAGSHIP VISUALS

- Segment revenue panel w/ is_disclosed shading
- Revenue-mix treemap
- Adoption-vs-revenue momentum
- CVE burden + severity stacked area
- Exploited-rate ranked bar
- Capstone scorecard scatter

■ METHODOLOGY

- 1 Vendor crosswalk: ticker ↔ CIK ↔ CPE strings — the hard entity-resolution artifact.
- 2 Rate-limited, cached ingest: NVD, KEV, EDGAR segment facts, survey CSVs.
- 3 Pillar A — segment panel + revenue mix; flag undisclosed cloud lines.
- 4 Pillar C — CVE attribution via CPE; KEV join; exploited-rate + lag.
- 5 Capstone — vendor scorecard; Spearman: growth vs. severe/exploited load.
- 6 pandera gates → Streamlit scorecard → pytest + CI → licensing/limitations README.

■ RISKS & HONESTY CONSTRAINTS

- Never fabricate Azure/GCP dollar revenue — MSFT/Google disclose growth % only; model the gap with an is_disclosed flag.
- CPE attribution is imperfect — counts are estimates; validate a sample, report attribution error.
- True market share is paywalled (Gartner/Canalys/Synergy) — press headlines are context only, never a dataset.
- Keep to ~10 vendors — expanding the crosswalk balloons entity-resolution effort; equity analysis stays non-causal.

Industry Value-Chain & the Margin-Distortion Anomaly

Conform ~8 of Damodaran's industry datasets onto one `dim_industry` spine in DuckDB, test one honest cross-dataset thesis, and export a Supabase-ready Postgres database.

DIFFICULTY

Advanced

Entity resolution + accounting literacy

EFFORT

~55–80 h

Reuses the portfolio plumbing

DATA CLEANLINESS

★★★★★

Clean values, dirty join key

UNIVERSE

94 industries

Jan-2026 · TTM ≈ Q3 2025

DELIVERABLE

Postgres DB

Runnable pipeline + Supabase SQL

■ THE INTERVIEW STORY

"I conformed ~30 of Damodaran's industry tables onto one `dim_industry` spine, then used the `join` to surface a claim no single page states — **EV/Sales tracks the R&D-adjusted margin, not the reported one**. Every number is validated against his firm counts, and I'm explicit these are value-weighted aggregates — association, not causation."

■ ONE SPINE: THE VALUE CHAIN

R&D

Intangible investment — intensity + Damodaran's capitalized-R&D estimate.

Margins

Reported vs. R&D/lease-adjusted operating margin — the distortion.

Financing

Dividends, buybacks, issuance, debt — the funding & payout signature.

Headline: EV/Sales tracks the adjusted margin better than reported (R^2 0.425 vs 0.406) across 84 non-financial industries; the adjustment reranks only 1/84 by ≥ 5 places (p 0.998) — the distortion is concentrated in the high-R&D tail.

■ KEY ANALYTICAL QUESTIONS

- 1 How far does the R&D/lease adjustment move each industry's margin, and **how many rank reversals** result?
- 2 Does the market (**EV/Sales**, **P/E**) track the reported or the adjusted margin?
- 3 Which industries return capital **funded by debt** (net buybacks + net debt raised)?
- 4 Do gross buybacks **overstate cash returned** where stock-based comp is high?
- 5 Do industries **cluster** into life-cycle / financing archetypes?
- 6 (panel) Is the reported-vs-adjusted gap **widening** across vintages?

Industry Value-Chain — pipeline, schema, risks

■ VERIFIED DATA SOURCES (DAMODARAN, NYU STERN)

SOURCE	ROLE	ACCESS
margin · R&D · finflows	Core thesis: margins, R&D intensity, financing flows	Direct /pc/datasets/<stem>.xls
psdata · pedata · divfcfe	Market payoff: EV/Sales, P/E; payout vs FCFE	Same host + 94-industry key
Archives	Prior vintages → the multi-year panel	△ /pc/archives/<stem><YY>.xls
SEC EDGAR (opt.)	Within-industry dispersion (extension)	Bridge via SIC — GICS is paywalled

■ FLAGSHIP VISUALS

- Reported-vs-adjusted margin dumbbell
- Rank-reversal bump chart
- Market-pricing scatter (EV/Sales)
- Financing quadrant (buybacks × debt)
- Life-cycle cluster map

■ METHODOLOGY

- 1 Ingest 6 canonical .xls (+2 local) → SHA-256 pinned vintage.
- 2 Parse — dynamic header (row-0 & row-7), NA→NULL, strip Total rows.
- 3 Resolve industry_id via alias crosswalk; **fail loud** on any unmatched name.
- 4 DuckDB star schema — dim_industry + 8 facts + wide mart.
- 5 Analysis — rank-reversal + market-pricing OLS/HC3 (numpy).
- 6 Export → Supabase Postgres (schema, seed, mart, RLS) + CSVs.

■ RISKS & HONESTY CONSTRAINTS

- Value-weighted aggregates (n≤94) — association, not causation; ecological fallacy.
- R&D-adjusted-margin circularity — never regress it on R&D; a lint blocks the pairing.
- Net Equity Change = issuance – buybacks (data over the file's wrong note).
- Join on a surrogate id, not the text — catches the real Healthcare typo; URLs overwrite each Jan → pin + hash.

The data foundation — free & adversarially verified

Every source behind the four projects passed **43 adversarial verification checks**. No paid data, no ToS-violating scrapes — and the failures found during verification are documented with working replacements.

✅ FULLY OPEN — NO ACCOUNT

- **SEC EDGAR XBRL** — 3 of 4 projects; User-Agent header, 10 req/s, US filers only.
- **NVD CVE API 2.0 · CISA KEV · MITRE cvelistV5** — the security record.
- **OWID COVID · Oxford OxCGRT** — cases & stringency (OWID frozen Aug 2024).
- **Damodaran NYU** — industry benchmarks (URLs rotate each January — pin).
- **yfinance** — all projects; fragile (429s) → fetch once, cache, retry.
- **Wikipedia · Stack Overflow survey · World Bank · Bain/Altgamma headlines · company URD PDFs · GH Archive.**

🔑 FREE WITH AN INSTANT KEY

- **FRED** — ~120 req/min; macro + VIX series.
- **BEA** — GDP-by-industry (optional).
- **BLS** — 500 queries/day (most series also on FRED).
- **Kaggle** — Fortune enrichment; pin slug + version.

🚫 PAYWALLED — EXCLUDED, DO NOT SCRAPE

- Statista · Macrotrends (ToS forbids bots) · stockanalysis.com.
- Gartner / Canalys / Synergy / IDC share reports → replaced by filings, free headlines, and revenue-mix / adoption proxies.

⚠️ CORRECTIONS FOUND IN VERIFICATION

- **Stooq direct CSV → blocked** by a JS anti-bot wall. Fix: yfinance / pandas-datareader readers, cached.
- **Fortune combined CSV → 404**. Fix: per-year files 2005–2019, concatenated, SHA-pinned.
- **Financial Modeling Prep → endpoints changed/capped**. Fix: SEC EDGAR XBRL as ground truth.

THE REUSABLE TOOLKIT

- **SEC EDGAR XBRL client** — used by 3 of 4 projects.
- **yfinance fetcher** with Parquet caching + retry — all four.
- **Damodaran .xls benchmark loader** — Luxury + Fortune 500.
- **Generic cached HTTP client** (retry/back-off, on-disk cache) — every REST/CSV source.

Defensible numbers, shared spine, compounding effort

ONE ENGINEERING SPINE ACROSS ALL FIVE PROJECTS

Free verified APIs

EDGAR, FRED, NVD/KEV, OWID, Damodaran...



Cached ingestion

Fetch once → Parquet; retry/back-off; pinned versions



DuckDB warehouse

raw → staging → marts, with lineage



Quality gates

pandera checks + pytest over every formula



Delivery

Streamlit dashboard + GitHub Actions CI + findings README

HONESTY AS A FEATURE

- Metrics are **defined where they're defensible** — “market share” means share within the covered, listed peer set, reproducible from primary filings.
- **Disclosure gaps are modeled, not papered over** — is_disclosed flags for Azure/GCP; attribution error reported for CVE↔vendor matching.
- **Crosswalks and caveats are first-class artifacts** — taxonomy mappings, alias tables, FX conventions, and frequency-alignment choices are all documented and tested.
- Each brief ships with **pre-written honest CV bullets** and clearly scoped extensions (backtests, event studies, factor decompositions, auto-refresh).

THE NARRATIVE ARC

- **Start: Fortune 500** — prove the data-engineering craft and build the shared toolkit (~40–70 h).
- **Flagship: Sector Rotation** — a formal event study with CARs and t-tests; the strongest quant story per hour invested.
- **Recruiter magnet: Luxury Comps** — sell-side comps work: normalization, crosswalks, valuation vs. fundamentals.
- **Capstone: Vendor Intel** — finance × cybersecurity; a measured growth-vs-risk trade-off nobody else in the stack of CVs will have.